

SentinelID Operational Protocol: Standard Operating Procedure (SOP) for Border Control Deployments

1. Purpose, Scope, and Strategic Intent

The security of the India–Nepal and India–Bhutan borders presents a unique operational challenge. Under bilateral treaty arrangements, these frontiers remain "open," allowing citizens to cross without traditional passports. This environment necessitates a shift from manual guesswork and visual inspection to AI-driven forensics. SentinelID is designed to bridge the gap between high-security international airports and remote Border Out Posts (BOPs), providing Sashastra Seema Bal (SSB) personnel with indigenous, cost-effective tools to identify suspicious travel documents, cross-border smuggling, and human trafficking threats within a largely undocumented flow of people. The primary objective of this protocol is to establish a standardized, offline-first workflow for document verification and fraud detection. By implementing this SOP, personnel can move from subjective visual checks to evidence-based screenings. This ensures that every document processed undergoes a rigorous, multi-layered validation sequence, reducing verification time from minutes to seconds. The scope of this document encompasses the end-to-end operational pipeline, including:

- Initial hardware readiness and environment configuration.
- Document ingestion and automated multi-engine extraction.
- Advanced forensic tampering analysis and biometric matching.
- Risk-based decision-making and the generation of a hash-chained investigative audit trail. Through these procedures, the SSB can bolster national security and maintain high traveler throughput while ensuring that every encounter is recorded with the forensic proof required for legal accountability under the DPDP Act 2023.

2. Field Readiness and System Configuration

In remote land-border environments, reliable internet connectivity is a luxury. Therefore, hardware-level preparedness is critical for "edge-inference"—the ability to run complex AI models locally without relying on cloud-based APIs. Proper configuration ensures that the system remains functional during network outages and provides the rapid response times necessary for field operations.

2.1 Required Hardware Checklist

The following hardware must be verified and operational before deployment begins:

- **NUC-class Edge Box:** The "System Brain" of the operation. This compact, high-performance computing unit runs OCR, tamper detection, and liveness processing locally to enable zero-connectivity operation.
- **Document Scanner/Webcam:** High-resolution capture devices for document ingestion and live facial images.
- **Mobile Kiosk/Tablet/Laptop:** The primary interface for the officer to interact with the SentinelID dashboard.
- **Local Storage Media:** For maintaining session caches and audit logs before synchronization.

2.2 Offline-First Configuration

To maintain network-independent operations, the system utilizes **Docker Compose** to orchestrate modular services (OCR, Tamper Detection, Face Verification) as a single, local ecosystem. **Steps for Initialization:**

1. **Service Orchestration:** Launch the Dockerized modules to ensure the backend framework (FastAPI) can coordinate verification processes without an external link.
2. **Database Initialization:** Initialize the local **PostgreSQL or SQLite** database. This local instance is essential for storing session data, document records, and the hash-chained audit log.
3. **Session Caching:** Verify that the **Redis** cache is active to provide fast session handling and temporary storage of results during the screening process.

3. Step 1: Document Ingestion and OCR/MRZ Extraction

The initial contact phase is critical. Strategic use of automated multi-engine Optical Character Recognition (OCR) reduces human error and significantly accelerates the extraction of biographical data. This step transforms raw image data into actionable, structured information for database validation.

3.1 Procedural Steps for Capture

1. **Document Placement:** Place the ID document on the scanner or under the webcam. Ensure the Machine Readable Zone (MRZ) is clearly visible.
2. **Live Capture:** Trigger the capture. The system will display an **animated OCR overlay**. This provides real-time feedback, allowing the officer to correct field misalignments or glare issues immediately before the forensic stage begins, significantly saving processing time.
3. **Preprocessing:** Before extraction, the system automatically applies **deskew, denoise, and contrast enhancement** filters. This is vital for maintaining high accuracy when processing documents that may be worn, damaged, or captured in low-light conditions.

3.2 Multi-Engine Extraction and Fuzzy Matching

SentinelID utilizes a redundant pipeline featuring **Tesseract OCR, EasyOCR, and PassportEye** for MRZ parsing. This multi-engine approach ensures reliability in extracting the following data points:

Data Category	Specific Fields Extracted
Biographical Data	Name (First, Surname), Nationality, Date of Birth (DOB), Gender
Document Data	Document Type, Document Number, Issuing Country, Expiration Date
MRZ Data	Full MRZ String, Checksum Validations (DOB, Number, and Expiration)

Fuzzy Watchlist Matching: Following extraction, the system performs **Fuzzy Matching** against local watchlists. This handles name transliteration variants common at the India-Nepal/Bhutan borders, ensuring that slight spelling differences do not allow high-risk individuals to bypass security.

4. Step 2: Forensic Analysis and XAI Heatmap Interpretation

The core of the SentinelID system is **Explainable AI (XAI)**. SentinelID provides transparent evidence through visual overlays and natural language summaries, allowing the officer to understand exactly *why* a document is flagged.

4.1 Interpreting the Tamper Heatmap

The system generates a **Tamper Heatmap** using **Error Level Analysis (ELA)** and **Grad-CAM** (Class Activation Mapping) via a fine-tuned MobileNetV2 CNN.

- **Grad-CAM Interpretation:** This proves the AI's focus area. It provides visual evidence of where the model is "looking" to reach its verdict. A red zone on a blank area may be a false positive, whereas a red zone on a date or name is a critical alert.
- **ELA Technical Logic:** ELA identifies "differing levels of JPEG compression." Since modified areas of an image are re-saved with different compression artifacts than the original, ELA highlights these digital splicing zones.
- **Photo Replacement:** High-intensity clusters around the portrait area indicate the photo may have been pasted over the original.
- **Altered Text:** "Pulsing" red boxes over dates (e.g., DOB or Expiration) suggest font inconsistencies or digital splicing artifacts.

4.2 Two-Factor Forensic Proof

To ensure accuracy, the system layers visual evidence with metadata analysis:

1. **Visual Tampering:** Detection of compression artifacts and noise inconsistencies through ELA.
2. **Metadata/EXIF Analysis:** Checking the image file's hidden data for signs of editing software or inconsistent timestamps.

4.3 Natural Language Summary

The officer must cross-reference the visual heatmap with the **LLM-generated summary**. For example: *"This passport shows signs of photo replacement (87% confidence) and a DOB inconsistent with the MRZ checksum."* This directs the officer to focus manual inspection on specific anomalies.

5. Step 3: Biometric Verification and Identity Graph Analysis

Identity verification requires confirming that the document holder is the rightful owner and that the person is not a repeat offender using multiple aliases.

5.1 Liveness and 1:1 Matching

To defeat "presentation attacks" (spoofing via printed photos or screens), the officer must initiate a **liveness check**.

- **Protocol:** The traveler is prompted to perform a simple action, such as a **blink or head-turn**.
- **Face Match:** Once liveness is confirmed, the system uses **ArcFace/DeepFace** embeddings to perform a 1:1 match between the live capture and the photo extracted from the ID document.

5.2 Identity Graph Analysis

SentinelID builds a **Cross-Scan Identity Graph** to catch sophisticated identity fraud:

- **Identity Clustering:** The system compares face embeddings against all previous scans stored in the **local PostgreSQL instance**.
- **Node-Edge Visualization:** If the same face appears under different names or nationalities across different scans, the system generates an alert.

- **Catching "Identity Rotators":** This strategically catches repeat offenders who attempt to bypass security by rotating forged documents while their biometric identity remains constant.

6. Step 4: Risk Score Evaluation and Command Decisions

All module outputs are fused into a single **Weighted Risk Score (0-100)** facilitated by an animated dial for rapid decision-making.

6.1 Decision Matrix

Risk Level, Score Range, Action Required

Low Risk (Green), 0 - 30, Automated Approval: Proceed with standard clearance.

Medium Risk (Yellow), 31 - 70, Manual Review: Flag for secondary inspection; review XAI heatmaps.

High Risk (Red), 71 - 100, Immediate Escalation: High probability of fraud; initiate detention protocols.

6.2 Command Mandates

- **OCR Confidence Threshold:** Officers **must not** use "Automated Approval" for any document where the OCR confidence score is below **85%**. Such cases require manual field verification to ensure extraction accuracy.
- **Evaluation of Score Breakdown:** A high score caused by an "OCR Confidence" issue might simply be a dirty document, whereas a high "Tamper Score" combined with a "Face Match" failure is a critical security alert.
- **Human-in-the-Loop:** Professional judgment must always validate "Review" and "Fail" verdicts.

7. Step 5: Investigative Audit Trail and Post-Encounter Documentation

Under the **DPDP Act 2023**, maintaining a secure and transparent record of identity processing is a legal mandate. SentinelID ensures accountability through a "hash-chained" audit trail.

7.1 Automated Logging and Hash-Chaining

Every action—including document upload, scan results, and the officer's final verdict—is timestamped and tied to a unique **Officer ID**. These logs are secured using a **SHA-256 hash-chained** structure, where each entry contains the hash of the previous record. This makes the log immutable and provides "Forensic Proof" for legal proceedings.

7.2 Documentation and Syncing

- **Investigation Report:** With one click, the officer can generate a **PDF Investigation Report** containing the extracted data, the tamper heatmap, the face match results, and the risk summary. This serves as a permanent evidence record.
- **Offline-to-Online Sync:** Once connectivity is restored, the local audit logs and identity graph data are automatically synced to the central SSB database, ensuring intelligence is shared across all checkpoints.

8. Contingency and Troubleshooting

8.1 Physical Wear vs. Digital Splicing

Physical damage, such as water stains or heavy creasing, can sometimes trigger a "Suspicious" verdict.

- **Protocol:** Officers must distinguish between **Physical Wear** (visible to the eye as surface damage, creases, or fading) and **Digital Splicing** (inconsistencies in JPEG compression detected by ELA that are invisible to the naked eye).
- **Manual Override:** If the AI flags a region corresponding only to physical wear, officers are authorized to use a manual override.

8.2 Hardware Failure Protocol

In the event of a total system failure (e.g., Edge Box malfunction):

1. **Manual Verification:** Revert to traditional manual document inspection and visual face matching.
2. **Manual Logging:** Maintain paper-based logs of all travelers.
3. **Digital Ingestion:** Once the system is restored, manually enter the paper logs to maintain the continuity of the digital audit trail.**Directives:** The safety of the officer and the security of the border remain the ultimate priority. Automated results should never supersede an officer's direct observation of immediate security threats.